



Picture of Lake Roland Dam, taken by AES Consulting

AES Consulting

Tarrodan Earthen Dam Program

Team Members: Eric O'Leary, Abigail Drager-Davidoff, Sean Burns

Faculty Advisor: Dr. Min Deng

Towson University

Table of Contents

1 Executive Summary	1
2 Introduction and Objectives	1
2.1 Background	1
2.2 Objectives	2
2.3 Key Metrics	2
3 Program Design	3
4 Financial and Economic Impact	4
4.1 Pricing and Costs	4
4.2 Stakeholder Analysis	6
5 Key Assumptions	7
6 Risk and Risk Mitigation	7
6.1 Key Risks	7
6.2 Risk Matrix	8
6.3 Sensitivity Analysis	9
7 Data and Data Limitations	9
7.1 Data Sources	9
7.2 Data Limitations	10
8 Conclusion and Recommendations	10
References	11
Appendices	13
Appendix A – Risk Management	13
Appendix B – Assumptions	14
Appendix C – Data Wrangling	16
C.1 Data Quality Chart	16
C.2 Preliminary Data Analysis	17
C.3 Data Imputation	17
Appendix D – Modeling	20
D.1 – Dam Repair Costs	20
D.2 – Climate Modeling	21
D.3 – Full Model	21

Appendix E – Inflation Analysis -----26

Appendix F – Stakeholder Analysis -----27

Appendix G – Financial Calculations -----29

 G.1 – Repair Bond -----29

 G.2 – Premium Calculations -----29

 G.3 – Labor Costs -----29

Appendix H – Sensitivity Analysis -----30

Appendix I – Future Investments -----33

1 Executive Summary

Tarrodan is at a crossroads. Aging dam infrastructure and accelerating climate change significantly increase the risk of dam failures, as seen in global news. Current projections call for over 1,950 expected failures in the next 10 years, before accounting for climate change. Each loss is expected to cost 400 million Qalkoons (Qm) for reconstruction and collateral damage. This has led to the need for an insurance program to protect against future economic and environmental losses.

AES Consulting has created an action plan detailing the creation of a program to provide disaster relief and minimize the risk of future dam failures by updating aging infrastructure. This plan reduces the probability of dam failure for thousands of dams, significantly reduces expected losses of human life and money, creates jobs, and helps prepare the nation for upcoming threats of overflow or failure.

Monitoring actual losses, reserves, and dam data updates allows the government to measure the success of the program. By adopting this program, Tarrodan is projected to decrease expected loss from dam failures by 73 billion Qalkoons (Qb) over the next ten years, while also fully covering all injuries from dam failures. These savings come from repairing insufficient dams or those with major risk to human life, reducing the probability of failure and the losses associated therein.

This program was designed to be conservative and flexible given the amount of funds being asked and the potential consequences to human lives and the country's economy. We have conducted robust modeling to minimize the chances of the program becoming insolvent and have taken great care to ensure that the program is funded equitably, taking contributions from all of Tarrodan and providing coverage to all of Tarrodan.

2 Introduction and Objectives

A sign of true leadership is preventing a disaster before it occurs rather than simply reacting to it afterwards. The council of Tarrodan has proven their excellence by forming the Earthen Dam Commission to stave off certain calamity. This action plan allows the citizens to sleep easier knowing that their infrastructure is both structurally sound and prepared for future shifts in the local and global climate. The proposed initiative invests in an inspection, maintenance, and disaster contingency insurance plan with the intent to save money, and more importantly, the lives of the Tarrodan people.

2.1 Background

Tarrodan is home to over 20,000 dams, of which more than half have not been surveyed by a structural engineer for soundness. Of those dams that have been analyzed, half are rated below satisfactory. Seventy percent of the largest dams – those that would lead to catastrophic damage in the event of failure – have not been surveyed. Each of the three major regions – Flumevale, Lyndrassia, and Navalidia – face unique challenges in geography, dam requirements, and potential

risks to overcome to ensure the safety and prosperity of the nation as a whole. The significant wealth disparity between these regions means that the cost of dam failure and the subsequent costs of recovery affect the regions disproportionately.

2.2 Objectives

AES Consulting has been tasked with developing an insurance program to mitigate the risk of dam deficiency, as well as restore the livelihoods of the people impacted by future dam failures. Each region will contribute towards an equitable insurance system to cover risks throughout the nation while ensuring that the fund remains solvent. This government funded program is designed to cover all the dams in the nation to ensure unbiased safety measures for all areas. We will outline the structure of our program, including proposals for funding, changes in inspection requirements and reporting, and financial incentives for preventative maintenance to mitigate both costs and the risk of failure.

This dam insurance and repair plan achieves five important objectives for the benefit of Tarrodan:

1. Reduce loss of lives from dam failure
2. Provide disaster relief and recuperate losses
3. Reduce frequency of failure
4. Predict and prepare for the effects of climate change
5. Equitably obtain funding

2.3 Key Metrics

While there are many factors that could be used to quantify the success of this infrastructure insurance program, AES Consulting has identified the three metrics that will be the most helpful: actual annual loss, data completeness, and reserve calculations.

Actual annual loss is calculated at the end of the fiscal year and measures the financial cost of Tarrodan's dam failures. As the infrastructure repairs are completed, the frequency of dam failures will decrease, leading to this metric decreasing every year, when adjusted for inflation.

Missing dam ratings will be updated as inspections are completed, with an expectation that they are all updated within five years to match the required standard of inspection frequency of once every five years. These inspections will also include a dam assessment rating, bridging the gap in the data and allowing the engineering planning division to correctly prioritize repairs.

Evaluating the program's reserve funds annually will help officials track the sufficiency of the premium amount based on the initial available data. This will naturally fluctuate as missing data is filled in and premiums are recalculated, but a reserve pool being too small or too large over the long term is a cause for re-evaluating the current plan structure.

3 Program Design

AES Consulting is proposing a program that is designed to both insure against and reduce the likelihood of failure of Tarrodan's dams. We plan to fully insure every dam in the country for property, liability, and economic losses. As we do not know which dams in Tarrodan are privately owned or privately insured, we assume we cover every dam to prevent underestimation. To reduce the likelihood of failure, we will also repair all dams that pose a high or significant hazard. This plan avoids the pitfalls of adverse selection by independently assessing the structural integrity of each dam rather than relying on reports from the dam owners. Additionally, because we are collecting premiums as a pooled tax, we remove the incentive misrepresent the failure risk of a dam to lower their premiums.

Our program will proceed in three phases: *Implement*, *Adjust*, and *Maintain*. The suggested timeline of the stages, as well as plan logistics, are detailed below.



Stage I: **Implement**. This program requires a reserve pool to fund the repair and renovation of existing dams. This reserve pool will be funded by issuing 10-year bonds to cover the projected costs of the repairs. This program also requires funding to cover claims, which will be maintained by annual premiums, collected as a regional tax that will be established in the first year. Each region will collect a total based on its GDP and will collect the tax from citizens via two income taxes, one on all residents, and a second on dam owners. This ensures poorer regions and individuals are not charged unmanageable sums to cover a program that is of public interest, and that individual owners are still held responsible.

This program requires the hiring of additional structural engineers. The funding for these engineers includes salaries, logistics such as office space, IT and support staff, and travel expenses. Subcontracting construction companies will begin as soon as the initial bond has been secured, focusing on high risk, poor or unsatisfactory dams that would maximize the reserve pool efficiency.

Stage II: **Adjust**. After the initial implementation period, all dams will have been scheduled for inspection. During this inspection process, the program will be adjusted to match the new data. A fraction of the dams that have not yet been repaired by the program will be selected and begin repairs. The dams will be selected for repair using the updated database, chosen from the set of dams that pose a high or significant hazard and ordered by expected reduction in losses minus the repair cost. It is likely that renovations will exceed time and monetary budgets (Petheram &

McMahon, 2019), so future repair estimates will be recalculated annually as well. As the dam database reaches 100% completion, the premium and reserve calculations can be updated to improve the model's accuracy from the initial predicted model forecast. Insurance payouts due to dam failure will be reviewed to ensure the premium reserves are within acceptable margins.

At the end of this stage, analysis on climate change and major model corrections will occur. Grants and programs to support the continued success of this plan will be implemented using excess collected monies. These programs include research into the safe decommissioning of inefficient dams, updating flood maps, structural requirement updates for climate change, and sediment analysis. For a summary of current research objectives, see Appendix I.

Stage III: Maintain. Once the calculations and schedules have been adjusted, the plan will move into the maintenance phase. We recommend following the industry standard of 5-year professional inspections by the government, with more frequent inspections for poorly assessed dams. Reserves will be monitored and audited to ensure that the plan continues to be solvent without overcharging taxpayers. We also suggest creating an online portal where dam owners can upload periodic 6-month visual inspection reports to identify problems early.

4 Financial and Economic Impact

4.1 Pricing and Costs

As this program is to be paid for by public funds and responsible for life-saving decisions, it must be financially responsible and transparent with its spending. The program was designed with this scrutiny in mind. We expect it to pay for itself in perpetuity while repairing a significant number of dams in need within the first ten years.

The following chart shows expected inflows and outflows for the next ten years of the program in millions of 2025 Qalkoons, except for the initial bond.

	INFLOW			OUTFLOW			MISC
	Premium	Interest	Repair Costs	Claims	Admin Costs	Bond Costs	Original Losses
2025	95207.29	2355.23	-871.65	-79754.72	-150.0	-1130.06	-82192.93
2026	92746.30	3141.92	-977.64	-78411.66	-50.0	-1097.15	-82452.93
2027	91254.68	3921.40	-954.80	-77472.51	-50.0	-1065.19	-82714.97
2028	90218.55	4701.38	-1124.63	-76661.21	-50.0	-1034.16	-82979.07
2029	89323.99	5479.42	-1361.27	-75965.04	-50.0	-1004.04	-83245.25
2030	88554.69	6255.43	-1684.56	-75307.32	-50.0	-974.80	-83513.53
2031	87832.45	7031.34	-1820.54	-74849.70	-50.0	-946.41	-83783.93
2032	87333.36	7811.13	-2018.82	-74484.35	-50.0	-918.84	-84056.46
2033	86935.61	8588.61	-2216.46	-74326.97	-50.0	-892.08	-84331.14
2034	86768.51	7799.55	-9086.65	-73616.97	-50.0	-21487.38	-84608.01

Table 1 – Program Inflows, Outflows, and expected financial losses (claims) without the program

The program will initially be funded by a 10-year bond. The total value of the bond will be set as the total estimated cost of the dam repairs plus 100Qm for additional administrative costs to establish the program. Calculations of repair costs are detailed in Appendix D.

This program has two inflows that are collected annually: premiums and interest. Premiums are set annually to cover administrative costs and the 95th percentile of expected losses for the year. To maintain equitability, premiums are collected from each regional government proportional to their percentage of the national GDP: Flumevale 52.0%, Navaldia 42.1%, and Lyndrassia 6.9%. This leads to respective first-year premium collections of 49,495, 40,049, and 5,662Qm respectively. Within each region, this premium is collected as a combination of an income tax on the citizens to cover government owned dams and an income or profit tax on the owners of dams so that individual landowners are protected from unaffordable taxes.

Navaldia and Lyndrassia each have ~8000 dams while Flumevale contains less than half that. But, the estimated loss per dam in Flumevale is 53.6Qm – significantly higher than Navaldia’s 35.7Qm, or Lyndrassia’s 32.0Qm. Flumevale has the smallest overall expected loss, but not by much. Collecting premiums by expected loss would have the wealthiest region paying the least, and the poorest region, Lyndrassia, paying almost as much as Navaldia, who has five times Lyndrassia’s wealth.

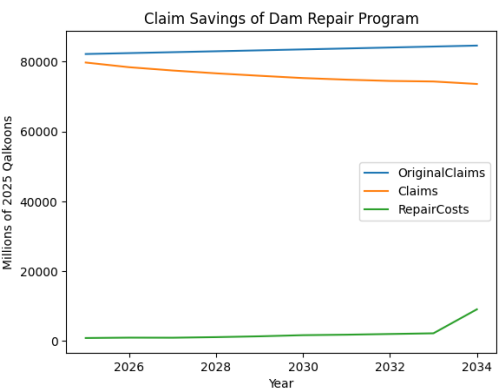
To reduce the effects of inflation and minimize the risks of insolvency, unused funds will be put into short-term investments, yielding the one-year spot rate. These investments must be low risk and be highly liquid to allow for payment of additional claims or over-budget repairs.

Outflows are divided into four main categories – administrative costs, bond costs, repair costs, and claims. Administrative costs include a one-time purchase of office space and equipment, as well as ongoing payments of salaries. Bond coupons and redemptions are another yearly outflow, with an upfront 1.07% underwriting fee (Joffe, 2015). In the final year of the program, the bond must be fully repaid. Repair costs are the total costs to repair the dams including planning, materials, and labor. The costs depend on the condition and height of the dam as described in (Association of State Dam Safety Officials, 2025). Complete details are available in Appendix D. Finally, claims are budgeted for by the database’s expected loss values and are paid out as necessary.

The largest and most variable outflow of this program comes from covering losses from dam failures, which varies between dams based on estimations of loss given failure. Any large-scale losses would also be distributed across the entire nation, rather than financially devastating a small area. This ensures that no party will face financial ruin from dam failure, as well as guaranteeing that dams will be repaired and rebuilt as necessary to maintain vital infrastructure.

After the program moves to the Maintain phase, we expect repair costs to shrink and the bond to be paid in full, decreasing outflows significantly. We expect claims to slowly increase due to climate change and the absence of the initial repair program. Premiums will be adjusted to match these new outflows using the now-completed assessment dataset.

This analysis is done translating each year into 2025 Qalkoons, with inflation naturally increasing many of these figures. Of particular interest, expected claim amounts increase over time when using raw numbers, whereas we see a decrease when adjusted to 2025 Qalkoons.



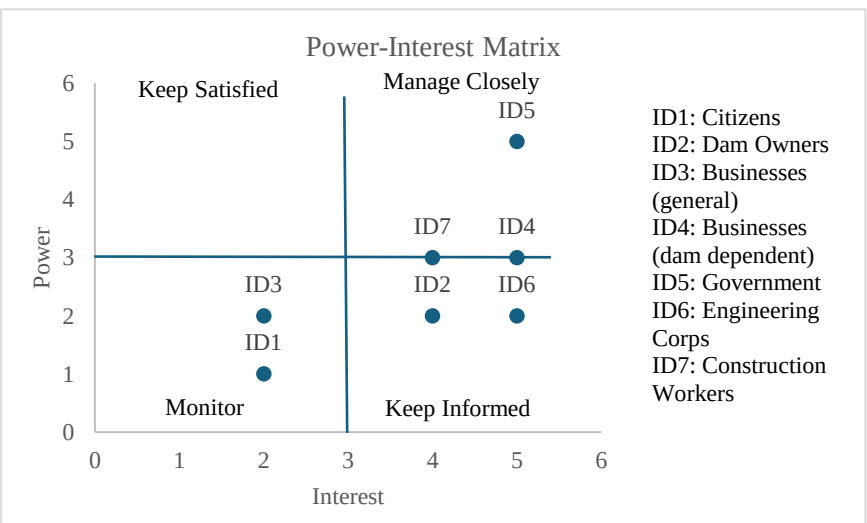
This program allows the government to provide an important service to its people. Without this program, losses would be significantly higher, with a difference of over 10Qb in 2034. While it is a major expense in relation to the national GDP, any dam failures are significant in relation to GDP and potentially lead to loss of lives as well. Tarrodan’s projected GDP in 2025 is 9,198.37Qb and expected total losses from dam failures is 79.75Qb. At the completion of the Adjust phase in 2035, the projected GDP increases to 13,135.11Qb whereas expected total losses from dam failures are reduced to 73.62Qb, solely because of the repair program. Note that these calculations all assume that the effects of climate change are constant and accurately modeled.

Collected premiums are sufficient to cover the 95th percentile of yearly claims, meaning that there is a 95% chance that the premiums will exceed the claims, resulting in a net positive cash flow. It is very unlikely that every year will have unusually high claims, and any years above that threshold will be covered by excesses from previous and future years. Short term government subsidies are an option should shortages occur early in the program. This, combined with the ability to adjust the premium model yearly, means that the program is extremely unlikely to become insolvent.

4.2 Stakeholder Analysis

To the right, there is a matrix which rates the influence and interest of the most important stakeholders for this program. We recommend the following strategies to properly manage each group:

The government will run the program and can do what



they see fit. General citizens and businesses should be monitored for general opinions on the program, especially reactions to the tax program and dam failure news. Dam owners and businesses who depend on dams should be kept more closely informed via sharing relevant information directly with these individuals. This information should also be made public for any other interested civilians. Relationships with the engineering corps and construction workers should be kept favorable as necessary, given how vital they are to the program's success.

5 Key Assumptions

Given the scope of such an undertaking, assumptions must be made to create an interpretable and reasonably accurate insurance model. The breakdown of the top six assumptions is detailed below, and a full list of assumptions can be found in Appendix B.

Metric	Assumption	Justification
Insurance Participation	100% of dams are insured under our program	Overestimation is preferred, and excesses can be invested. Additionally, the government may mandate this insurance for all dams
Dam Renovation Costs	Necessity and type of renovation depend on age and assessment level, cost depends on height	Estimations and justification are from (Association of State Dam Safety Officials, 2025). Further explanation in Appendix D
Average Dam Renovation Time	3 months	A US dam in Nevada of above average size was renovated in around 3 months (Bureau of Indian Affairs, n.d.)
Dam Improvement	Renovations improve a dam to satisfactory and decrease the failure rate to regional satisfactory average	It is undesirable to upgrade a dam to anything except satisfactory. For simplicity, we assume the dam performs averagely for dams of similar assessment level within the region
Climate Change Failure Rate Increase	Failure rate change is multiplicative, depending on assessment level	Multiplicative change is simple, and dam assessment level indicates ability to withstand non-ideal conditions. Details in Appendix D
Dam Failure per 10 Years	A dam can only fail once in ten years	Besides being highly unlikely (<.1%), rebuilding a dam takes time. A newly rebuilt dam will have a very low chance of failure

6 Risk and Risk Mitigation

6.1 Key Risks

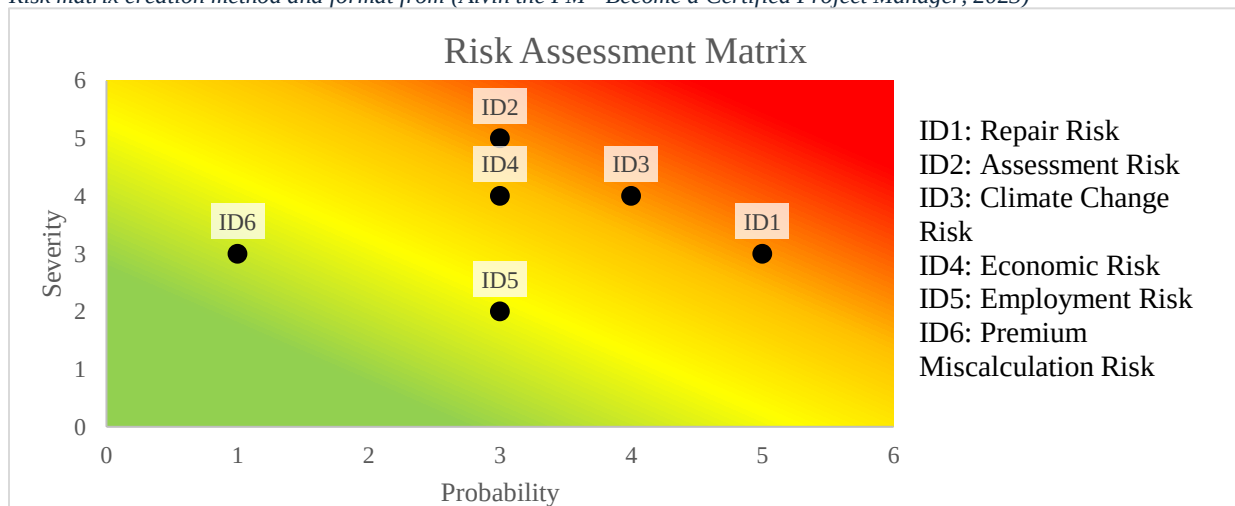
As with all endeavors, implementing an infrastructure program comes with risks that will need to be addressed and mitigated. We have included the top six risks here, with the full breakdown in Appendix A. The lesser risks should still be mitigated when the benefits outweigh the costs, but the major factors above should be incorporated into any decision-making process.

Risk	Description	Probability	Severity	Mitigation
------	-------------	-------------	----------	------------

Repairs go over cost and/or time budget	Construction delays and cost overruns impact the funding pool	Very High	Medium	Review similar issues that have happened in other parts of the world and budget accordingly
Uninspected dams have a different assessment distribution	Dams that are currently unrated are in worse condition than anticipated	Medium	Very High	Shift inspectors from other areas initially to survey these dams and get the national database up to current
Climate change increases dam failure rate	Rising water levels increase at a more rapid pace than expected, leading to failures	High	High	Work with engineering corps to identify possible new height requirements in higher risk areas, then reinforce existing dams accordingly
Economic downturn	Changes in economic conditions effect solvency	Medium	High	Public awareness campaigns to reinforce the cost/benefit analysis to the public & politicians
Lack of qualified structural engineers	Inspectors require an engineering degree and years of experience	Medium	Low	Incentivize engineering firms to take on associate engineers in anticipation of expanding the talent pool
Premium calculations underfund actual requirements	Missing data or unexpected circumstances may make the model inaccurate	Low	Medium	The premium amount will need to be recalculated during the adjustment period once the missing data has been collected.

6.2 Risk Matrix

Risk matrix creation method and format from (Alvin the PM - Become a Certified Project Manager, 2023)



6.3 Sensitivity Analysis

The following values are in millions of Qalkoons, adjusted to 2025 prices.

	WORST				EXPECTED				BEST			
INFLATION RATE	9.331%				3%				1.566%			
<i>ENDING FUNDS</i>	94,380Qm				135,918Qm				148,654Qm			
INTEREST RATE	0%				5.9%				12%			
<i>ENDING FUNDS</i>	85,791Qm				135,918Qm				214,672Qm			
REPAIR COST	2.2x				1x				-			
<i>ENDING FUNDS</i>	108,432Qm				135,918Qm				-			
RATE PER ASSESSMENT*	S	F	P	U	S	F	P	U	S	F	P	U
CLIMATE CHANGE	1	1.002	1.01	1.02	1	1.001	1.005	1.01	1	1.0005	1.0025	1.005
<i>ENDING FUNDS</i>	128,360Qm				135,918Qm				139,586Qm			

*S = Satisfactory, F = Fair, P = Poor, U = Unsatisfactory dam assessment rating

Climate change is modeled by multiplying the probability of failure for a given dam by a constant based on its assessment, Satisfactory, Fair, Poor, or Unsatisfactory. These ratings describe how well a dam performs under non-standard conditions, which climate change slowly creates. Because these factors are unknown and unmeasurable, sensitivity analysis involving climate change assumes claims increase but premiums do not. For full details for climate change modeling, see Appendix D. For full details of sensitivity analysis, see Appendix H.

7 Data and Data Limitations

7.1 Data Sources

Data comes from four main sources:

1. The data provided, including the dam database, Tarrodan's financial information, and the Tarrodan encyclopedia entry
2. The US Bureau of Labor Statistics, which provided historical US inflation rates (Bureau of Labor Statistics, n.d.)

7.2 Data Limitations

Limitation	Description	Impact	Management
Missing Values	High proportions of missing values in many categories	Assumptions and imputation methods are not necessarily accurate. Missing data due to a lack of inspections may be worse than assumed	MICE is used to impute missing data. See Appendix C.
Relational Data	Data describing the similarity or proximity of dams to each other is not present	Comparing dams or creating summaries of types of dams is not possible with any degree of precision, making estimates very vague	Strategies that use this data are not used, and dams are grouped by region if at all
Insurance/Ownership Data	Data describing ownership (private, public) and current insurance policies (private, state, none) is unavailable	It is not possible to tell which dams must be insured and which can be left out of the plan	To provide a conservative estimate, we assume this program insures all dams
GDP Data	Only 5 years of GDP data provided	A small dataset including the outlier of COVID is unreliable for forecasting	Growth rate was calculated without COVID, noted to be unreliable

These are the limitations with the largest influence on the program. A full list of data limitations can be found in Appendix C.

8 Conclusion and Recommendations

This program will save Tarrodan an expected 73,027Qm in claims over the next ten years, with more expected beyond. Furthermore, dams will also be future proofed against climate change and properly inspected to catch problems earlier, replacing massive expected losses on failure with significantly smaller expected repair costs. This program can be improved by analyzing weather and location data to create stronger estimates. Also, as climate change continues to evolve and be better understood, changes in dam failure rates can be made more accurate over time. Overall, this plan will improve the peace of mind, well-being, and economic prosperity of Tarrodan and its people.

References

- Alvin the PM - Become a Certified Project Manager (Director). (2023, May 21). *Risk Assessment Matrix in Excel in 15 Minutes!* [Video recording].
https://www.youtube.com/watch?v=8k6P_g_XG84
- Association of State Dam Safety Officials. (2025, March). *The Cost of Rehabilitating Dams in the U.S.* <https://damsafety-prod.s3.amazonaws.com/s3fs-public/files/2025%20ASDSO%20Costs%20of%20Dam%20Rehab%20Report.pdf>
- Banham, R. (2024, February 13). *The Impact of Climate Change on Flood Risk Management.* <https://www.rmmagazine.com/articles/article/2024/02/13/the-impact-of-climate-change-on-flood-risk-management>
- Bureau of Indian Affairs. (n.d.). *Weber Dam Response | Indian Affairs*. Retrieved February 25, 2025, from <https://www.bia.gov/WeberDamResponse>
- Bureau of Labor Statistics. (n.d.). *United States Historical Inflation Rate* [Dataset].
https://data.bls.gov/timeseries/CUUR0000SA0L1E?output_view=pct_12mths
- Bureau of Labor Statistics. (2024, December 17). *Employer Costs for Employee Compensation—September 2024*. <https://www.bls.gov/news.release/pdf/ecec.pdf>
- Concha Larrauri, P., & Lall, U. (n.d.). *ASSESSING THE EXPOSURE OF CRITICAL INFRASTRUCTURE AND OTHER ASSETS TO THE CLIMATE INDUCED FAILURE OF AGING DAMS IN THE U.S.* Columbia Water Center.
<https://water.columbia.edu/content/assessing-risks-dam-failure-united-states>
- Donders, A. R. T., van der Heijden, G. J. M. G., Stijnen, T., & Moons, K. G. M. (2006). Review: A gentle introduction to imputation of missing values. *Journal of Clinical Epidemiology*, 59(10), 1087–1091. <https://doi.org/10.1016/j.jclinepi.2006.01.014>

Federal Reserve Bank of New York. (2025, February 28). *Center for Microeconomic Data | Inflation Expectations*.

<https://www.newyorkfed.org/microeconomics/topics/inflation.html>

Ho, M., Lall, U., Allaire, M., Devineni, N., Kwon, H. H., Pal, I., Raff, D., & Wegner, D. (2017).

The future role of dams in the United States of America. *Water Resources Research*, 53(2), 982–998. <https://doi.org/10.1002/2016WR019905>

Joffe, M. (2015). *Doubly Bound—The Cost of Issuing Municipal Bonds*.

https://belonging.berkeley.edu/sites/default/files/haasinstituterefundamerica_doublybound_cost_of_issuingbonds_publish.pdf

Petheram, C., & McMahon, T. A. (2019). Dams, dam costs and damnable cost overruns. *Journal of Hydrology X*, 3, 100026. <https://doi.org/10.1016/j.hydroa.2019.100026>

van Buuren, S. (2018). *Flexible Imputation of Missing Data* (2nd ed.). CRC Press.

<https://stefvanbuuren.name/fimd/>

Vink, G., & van Buuren, S. (n.d.). *miceVignettes*. miceVignettes. Retrieved February 10, 2025, from <https://www.gerkovink.com/miceVignettes/>

Appendices

Appendix A – Risk Management

Risk Category	Risk Subcategory	Risk	Definition	Mitigation
Operational	Human resources	Ability to recruit/retain	Ability to recruit and retain qualified engineers	Education incentives and work visa programs
Operational	Disaster	Increased natural disaster frequency	Hurricanes and/or other weather changes lead to increased flooding	Adequate spillways and dam renovations
Operational	Data Security	Data breach	Ransomware attack on the local or federal government database	Proper cybersecurity protocols
Strategic	Adverse Selection	Regional financial burden	The financial climate of a region may disproportionately change	Annual economic reviews
Strategic	Economic	Unexpected changes in economy	Changes to interest rates, inflation, and other factors	Annual economic reviews
Strategic	Political	Unexpected changes in political climate	Changes in majority political opinion about program necessity	Committee presentations on program effectiveness
Strategic	Government	Ineffective project implementation	Data from inspections do not properly transfer to the main database, requiring more manual inputs	IT data updates compared to engineer report frequency
Financial	Liquidity	Unexpected changes in liquidity	Revenue/expenses not matching expectation	Annual revenue stream audits
Financial	Fraud	Internal or contractor fraud	Funds for disaster relief or dam maintenance may be diverted	Annual revenue stream audits

Research has shown that cost and time overruns are very common in other countries that have implemented similar maintenance projects, running from 50-100% of the quoted repairs (Petheram & McMahon, 2019). This will not only affect the funding pool for the project, but the availability

of the engineers involved in the repairs. We recommend that the budget be adjusted to account for these increased costs that will inevitably occur.

One key assumption is that the overall condition for the dams that have not been inspected or otherwise not been rated for structural integrity is that they have a similar distribution of satisfactory, fair, poor, and unsatisfactory percentages. It is quite possible that the lack of oversight with these dams has left them in worse condition than those that have been consistently inspected. It is imperative that these dams be investigated for possible issues as soon as possible. We recommend shifting engineers off routine inspections during the first year of the program to update the national database so that future repairs can be prioritized correctly to mitigate the most disaster risk.

Rising water levels due to climate change have been an issue globally for dams and should be taken into consideration with the insurance program. Climate change has progressed more quickly than anticipated in several studies, and it is possible that it may escalate further in the near or long term. The findings from the engineering corps should be weighed heavily into decision making for budget and scheduling, as increasing dam heights as part of the maintenance process could increase the time and costs of these repairs.

The ebbs and flows of the local and global economy will inevitably have an impact on the popularity of infrastructure investment and insurance costs. It is imperative that future government compositions be aware of the savings of this program, both in terms of finances and lives. Cutting back on inspections or renovations may save money in the short term, but the financial report above clearly shows the long-term benefits of maintenance and insurance for your infrastructure.

Our program design calls for an influx of engineering talent to handle the increased number of inspections and construction oversight. This talent pool will also need to be monitored to account for sufficient junior inspectors to fulfill the inspection needs as well as covering for the inevitable retirement of senior engineers. It is possible that Tarrodan may need to import talent from other areas on work visas in the short term to meet the initial demand for engineering talent. Industry demand inevitably creates education opportunities, so we do not anticipate this to be a long-term shortage.

The program design calls for a large influx of starting capital in the form of bonds. Similar international proposals tend to be approved without major roadblocks, but it is always possible that bond vote fails to pass, which would require an alternative source of funding. The insurance program would not have a major impact on the overall government spending, so it would be feasible to adjust the budget accordingly if need be.

Appendix B – Assumptions

Metric	Assumption	Reasoning
--------	------------	-----------

General Assumptions		
Labor Hiring	We will be able to hire as many employees as necessary, they will have no issues traveling to location	The size and relative wealth of Tarrodan suggests that the talent pool will be sufficient.
Assessment Ability	Structural engineers will provide assessments during required inspections	Assessments are not likely to require large scale operations for most dams
Data Assumptions		
Height	Zero is missing value	Since height is measured from bottom of excavation to top, height of zero is entirely flat
Loss Given Failure Business	Missing value is not zero	If dams are in more remote locations, it may be difficult to judge business impact. This value may also indicate less than 1Qm, which is not negligible when summed
Distance to City	Zero is not a missing value	Dams may exist within city limits
Last Inspection	Missing value means no inspection since 2003	A paper trail should exist for any inspection, but 2003 is the earliest date and previous information may be stored elsewhere
Years Modified	Missing value means the dam has never been modified	There is no data available to suggest otherwise
Data Accuracy	Data, especially the probability of failure, is accurate	Those who modeled the data are likely accurate, and there is insufficient information to recompute
Population Growth Rate	Annual population continues to increase at 1% rate	Consistency in the previous 5-year census tallies suggest a steady growth rate.
GDP Growth Rate	6.74%	An average growth rate over time, which was then used for the entire period as a conservative estimate
Inflation	3%	Comparison suggests Tarrodan's rate is similar to the US rate. See Appendix E

Labor Costs	50Qm yearly operating budget	See Appendix G.
Dam Independence	The chance of failure for each dam is independent	The independence assumption was given by the documentation from Tarrodan
Dam Types to Cover	All dam types will have coverage provided	Tarrodan's provided data includes non-earth dams, despite the project title only referencing earthen dams

Appendix C – Data Wrangling

C.1 Data Quality Chart

Column Name	Type/Structure	Data Quality
ID	SOAD12345	No duplicates found
Region	Flumevale, Lyndrassia, Navalidia	CLEAN
Regulated	yes, no	CLEAN
Purpose	Recreation, Water Supply, ...	1184 (5.69%) missing values
Type	Earth, Concrete, ...	257 (1.24%) missing values
Height	12.345	24 (.12%) zeroes
Length	1.234	2671 (12.84%) missing values
Volume	12345	9678 (46.52%) missing values
YearCompleted	Yyyy	1384 (6.65%) missing values
YearsModified	yyyy with optional letter code after, can have multiple	1284 (6.17%) missing values
Surface	1.23456	2798 (13.45%) missing values
Drainage	1.23456	2463 (11.84%) missing values
Spillway	Controlled, Uncontrolled	12786 (61.45%) missing values
LastInspection	dd/mm/yyyy	10025 (48.18%) missing values, contains a date with year 2031
InspectionFreq	1.0	8116 (39.01%) missing values
DistanceToCity	1.23	10229 (49.16%) missing values
Hazard	Significant, High, Low	CLEAN
Assessment	Satisfactory, Fair, ...	2537 (12.19%) missing values
Assessment Date	dd/mm/yyyy	9773 (46.97%) missing values
ProbFailure	0.1234	CLEAN
Loss Given Failure – Property	123.4	7 (.03%) missing values

Loss Given Failure – Liability	123.4	12 (.06%) missing values
Loss Given Failure – Business	12.3	10730 (51.57%) missing values

C.2 Preliminary Data Analysis

A generic set of frequency tables, categorized by different variables of interest, reveals a few interesting pieces of data. Firstly, Flumevale has the smallest number of dams, at around 3500 dams or 17% of the nation’s dams. Secondly, Flumevale has no missing values for assessment and Navalidia has one, but Lyndrassia has 2500. Next, Flumevale also has the highest proportion of “satisfactory” assessments whereas Lyndrassia and Navalidia have “not rated” as the majority assessment. Of those that are rated, Navalidia has a disproportionate number of dams rated “fair” and “poor,” whereas Lyndrassia has a high proportion of missing values, though both have many listed as “not rated.” From this, we expect two major challenges to be updating Navalidia’s infrastructure and Lyndrassia’s data collection process.

One key limitation of the provided data is the high amounts of missing values across a variety of columns. Eight of the variables contain a minimum of roughly 40% missing data. In particular, around 30% of the dams did not have a last inspection date or an inspection frequency, implying they may not be inspected at all, or at least since the earliest listed date of 2003. Also, around half of the dams did not have a predicted loss given failure for business costs, making accurate estimation difficult. It is unclear if this implies that the dam has less than 1Qm expected business loss, as that is the minimum of the column. However, some dams with missing loss given failure are located in a city (distance to city of zero), meaning that their failure is highly unlikely to have minimal impact on business. As such, we assume these values are difficult to calculate, rather than negligible.

While this dataset does have several outliers, removing them would be detrimental. Most distributions are skewed right, so handling outliers would primarily remove the most important dams. As such, no data points are removed.

C.3 Data Imputation

Accurately estimating missing data has become significantly easier in the modern day. While simple imputation methods such as mean replacement work if the missingness of the data is independent of other variables, multiple imputation can sidestep that issue while providing unbiased estimates (Donders et al., 2006). Of the methods for multiple imputation, the industry standard is multivariate imputation by chained equations, or mice. For insights into the methodology and how to best utilize these methods, see (van Buuren, 2018; Vink & van Buuren, n.d.).

Note that as it is difficult, if not impossible, to determine the accuracy of imputed data without knowing the actual answer for the missing value, we trust in the research saying that these methods are accurate.

When working with the dataset, a problem emerges when using the best performing imputation method for each column – unsolvable equations. Certain numeric columns, such as height, length, volume, and loss given failure, are not able to be predicted via the standard method. This error often means that a column is a linear combination of some other columns, creating a non-solvable system of equations. However, in this case, the error message is misleading, as the problem is not with the method of imputation or the structure of the data, but the scale of the data. Specifically, variables such as volume range into the tens of thousands, while others such as probability of failure are less than one.

This disparity can be solved by standardizing each variable, performing the imputation, and then de-standardizing each variable. This method may not be ideal because it fails to account for missing values when computing means and standard deviations and may therefore introduce bias. As well, this method is not guaranteed to produce realistic values, such as producing a negative value for loss given failure or height.

Now, we need to conduct the imputation. As we will see in Appendix D, the only columns that are relevant to modeling loss and repair costs are region, height, assessment, probability of failure, and the three loss given failures. Of these, region and probability of failure are complete columns, so there is no need to impute them. Loss given failure can be simply imputed as a numeric value, using the default method of predictive mean matching. Assessment requires some more work. The methodology used to compute repair costs replaced assessments that were missing, unrated, or unknown with a sample from the United States' known assessments to maintain a similar proportion (Association of State Dam Safety Officials, 2025). However, as we are using imputation to replace other data, we see no reason to not use imputation for this as well, given it is likely to have higher accuracy. As such, we set all assessments that are not rated, not available, or missing as missing, and will therefore be imputed.

Do note that we use `quickpred()`. This function determines which columns should be used to impute other columns based on correlation, with a default minimum threshold of 0.1 correlation. This helps to speed up the process – which is key for a dataset of 20,000 rows and 23 columns – as well as make the relationship between variables slightly stronger.

```
# Standardization to avoid unsolvable equations
tempdam <- damdata |> mutate(across(where(is.numeric), ~(scale(.) %>%
as.vector)))

damstats <- damdata |>
  summarize_if(is.numeric, list(mean = mean, sd = sd), na.rm=T)
```

```

damimpprep <- tempdam |>
  mutate(Assessment = ifelse(Assessment %in% c("Unsatisfactory", "Poor",
"Fair",
                                "Satisfactory"),
                             Assessment,
                             NA)) |>
  mutate(Assessment = factor(Assessment,
                             levels = c(1,2,3,4),
                             labels=c("Unsatisfactory", "Poor", "Fair",
"                                Satisfactory"))) |>
  select(-c("LastInspection", "AssessmentDate"))

# No colinear columns, which is desirable
mice::find.collinear(damimpprep)

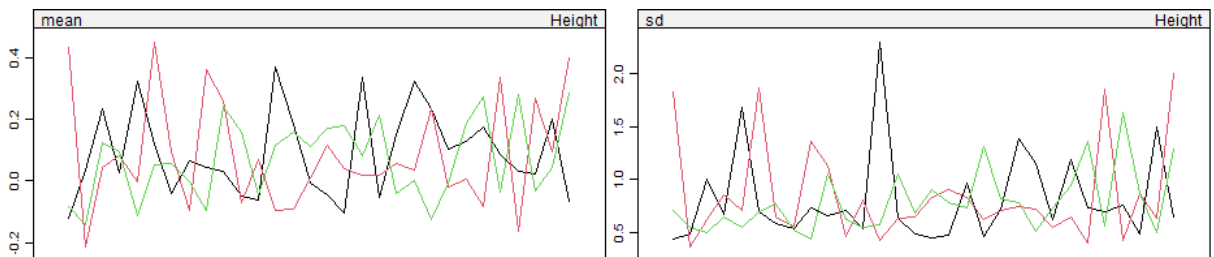
# Quickpred for faster calculation
init <- mice(damimpprep, pred=quickpred(damimpprep), maxit=0)

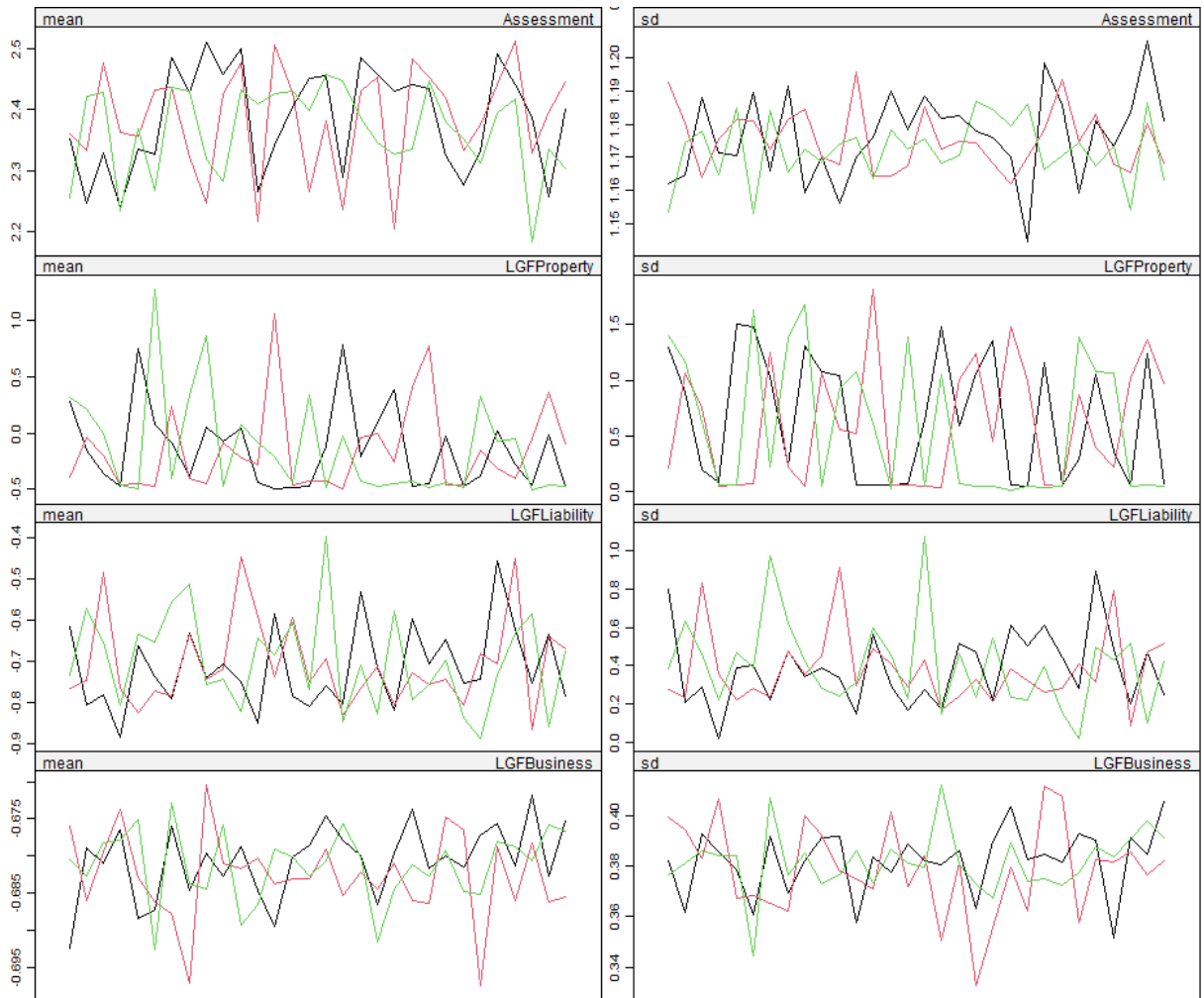
pred <- init$predictorMatrix
# Do not impute on columns of no interest
pred[1:5, ] = 0
pred[7:16, ] = 0

impute_cols <- mice(damimpprep, pred=pred, m=3, maxit=30)

```

After imputation, we check for a few markers of quality. Firstly, we examine the convergence of the algorithm. Ideally, across all imputations, as iteration number increases, there should be no notable trend found. As we see in the below images, across three separate imputations and thirty iterations, there is no noticeable trend in mean or standard deviation.





Next, we look for unreasonable values. Luckily, we do not have any– all height values are positive, though perhaps small with a minimum of .346 meters; all loss given failure amounts are within the previous minimums and maximums; and assessment cannot be anything besides its specified categories.

From this, we are comfortable concluding that our imputation has produced a reasonably accurate approximation of the missing data in the columns of interest. We can now use this data to produce a model with better performance than simpler methods, such as dropping rows or replacing missing values with column means or zeros.

Appendix D – Modeling

D.1 – Dam Repair Costs

In order to calculate dam renovation costs, we rely heavily on a 2025 study by the Association of State Dam Safety Officials, which describes a method for estimating repair costs using past data across all of the United States (Association of State Dam Safety Officials, 2025). This study uses

three metrics to determine the repair cost – height, age, and assessment. Generally, the acceptable level of assessment becomes stricter as age increases, and the cost to repair also increases as age increases. Within these groups, the height of the dam determines the specific estimated cost, with taller dams being more expensive.

Using these repair costs, all dams that require repairs are easily identified. Of these dams, we then choose which of Tarrodan's dams to repair by determining which are either profitable to repair or propose a high or significant hazard on failure. This reduces the significant economic impact of the program as much as possible while also minimizing the loss of human life. However, we must decide how to determine if repairing a dam is profitable. To do so, we define this as a positive amount of expected savings over the next ten years. Thus, we need to calculate how much we expect the repair to save. In order to do this, we need to know how a dam changes after it is repaired.

We assume the following: when a dam is repaired, its assessment is moved to satisfactory – why would we repair a dam to any other rating – and the probability of failure for that dam is adjusted to the minimum of its current probability and the regional average probability for dams originally rated satisfactory. This change in probability of failure allows us to calculate the expected loss over a ten-year period assuming repairs have been completed, which we can then compare to the previous expected loss.

D.2 – Climate Modeling

Unfortunately, the effect of climate change on the probability of dam failure is not well studied, and a professional, engineering-based estimate is not available. Therefore, we construct a modeling technique through logic and guesswork. The main piece of information we will use is the assessment of dams – satisfactory, fair, poor, or unsatisfactory. These are standardized terms which define under what circumstances a dam safety concern would occur. For example, we assume that a dam with a satisfactory assessment – no safety deficiencies and acceptable performance under all reasonable circumstances – will not have a noticeable change because of climate change, but an unsatisfactory dam – known dam safety deficiency – will suffer the most significant increase in probability of failure.

Using these assessments, we then define a multiplicative factor which we use to adjust the probability of failure after each year. The specific factor increases from satisfactory assessment to unsatisfactory and is generally less than 1% increase per year. Later, we will vary this factor for sensitivity analysis.

D.3 – Full Model

The data provided includes a probability of failure over the next ten-year period. However, we wish to model the dams on a yearly basis, as it allows for calculating yearly inflow requirements and the repair program's savings per year, rather than over ten years. In order to do this, we assume that the probability of dam failure is modeled by an exponential distribution, as it is the most common method to model continuous lifetime. Given this assumption, the provided probability of

failure is now equal to $F(10)$, the cumulative probability of failure within ten years. We can now solve for λ , the parameter of the distribution, which can then be used to calculate $F(1)$, the cumulative probability of failure within one year. As the exponential distribution is memoryless, the probability of failure in year two given that the dam survived year one is the same as the probability of failure in year one. Thus, we can model the probability of failure within one year and reuse it for all ten years of the repair program.

However, we do want to adjust these probabilities for climate change as per Appendix D.2 and dam repair as per D.1. For climate change, we multiply the λ for each dam by the climate change factor every year, according to the dam's assessment. For repair, once the dam is flagged as repair in a specific year, its assessment is changed to satisfactory for the rest of the ten-year period, and its probability of failure is adjusted to be the minimum of its own probability of failure and the average for satisfactory rated dams of the region. This is the best estimate for improving the dam's safety via repair possible due to the limits of the data.

Putting the previous sections of this appendix together, as well as the end of Appendix C Appendix C – Data Wrangling, we construct the complete model.

Firstly, we use MICE to impute missing data for height, assessment, and all three categories of loss given failure. Secondly, we calculate the estimated repair cost for all dams. These are then used to calculate expected savings after repair. At this stage, we also flag which dams are profitable or necessary to repair. Finally, we model each year – probability of failure, expected yearly loss, dams repaired, expected yearly savings, and more.

```
def simulate_1_year(
    year_start_funds_m_q: float,
    annual_admin_costs_m_q: float,
    dams: pd.DataFrame,
    control_dams: pd.DataFrame,
    repair_cost_matrix: np.ndarray,
    years_left: int,
    premium_z: float,
    claim_z: float,
    bond_total_m_q: float,
    bond_interest_rate: float,
    interest_1_year: float,
    inflation_rate: float,
    predicted_climate_effects: Dict[str, float],
    true_climate_effects: Dict[str, float],
    setup_cost_m_q: float,
    close_bond: bool,
) -> Dict[str, Any]:
    # Make copies to avoid overwriting past data
    repair_cost_matrix = repair_cost_matrix.copy()
    dams = dams.copy()
    control_dams = control_dams.copy()

    # 0. Start
    funds = year_start_funds_m_q
```

```

# 1. Charge Premium (95% loss) + Administrative costs
high_95 = compute_claims_at_z(dams, premium_z)
premium = high_95 + annual_admin_costs_m_q
funds += premium

# 2. Repair Dams
repair_stats = repair_dams(repair_cost_matrix, dams, years_left)
dams = repair_stats["Dams"]
funds -= repair_stats["RepairCosts"]

# 3. Pay Expected Claim Costs
dams = update_annual_claim_fields(dams)
claims = compute_claims_at_z(dams, claim_z, use_hidden=True)
control_claims = compute_claims_at_z(control_dams,
claim_z, use_hidden=True)
funds -= claims

# 4. Pay Admin Costs
paid_admin_costs = annual_admin_costs_m_q + setup_cost_m_q
funds -= paid_admin_costs

# 5. Pay Bond Costs
bond_payment = bond_total_m_q * bond_interest_rate
if close_bond:
    bond_payment += bond_total_m_q
funds -= bond_payment

# 6. Interest income
interest_income = max(interest_1_year * funds, 0)
funds += interest_income

# Advance time
dams["Age"] += 1
control_dams["Age"] += 1

# Apply inflation costs for next year
dams = apply_inflation(inflation_rate, dams)
control_dams = apply_inflation(inflation_rate, control_dams)
repair_cost_matrix *= (1 + inflation_rate)
annual_admin_costs_m_q *= (1 + inflation_rate)

# Apply climate factors
dams = apply_climate_effects(predicted_climate_effects,
true_climate_effects, dams)
control_dams = apply_climate_effects(predicted_climate_effects,
true_climate_effects, control_dams)

return {
    "Dams": dams,
    "ControlDams": control_dams,
    "EndFunds": funds,
    "Premium": premium,
    "RepairCosts": repair_stats["RepairCosts"],
    "DamsRepaired": repair_stats["DamsRepaired"],
    "HazDamsRepaired": repair_stats["HazDamsRepaired"],
    "Claims": claims,
    "OriginalClaims": control_claims,

```

```

        "AdminCosts": paid_admin_costs,
        "NewAdminCosts": annual_admin_costs_m_q,
        "BondCosts": bond_payment,
        "InterestIncome": interest_income,
        "RepairCostMatrix": repair_cost_matrix,
    }

def update_final_results(final_results: pd.DataFrame, year: int,
starting_funds: float,
                        year_results: Dict[str, Any]) -> pd.DataFrame:
    final_results.loc[year, "StartFunds"] = starting_funds
    final_results.loc[year, "Premium"] = year_results["Premium"]
    final_results.loc[year, "RepairCosts"] = -year_results["RepairCosts"]
    final_results.loc[year, "DamsRepaired"] = year_results["DamsRepaired"]
    final_results.loc[year, "HazDamsRepaired"] =
year_results["HazDamsRepaired"]
    final_results.loc[year, "Claims"] = -year_results["Claims"]
    final_results.loc[year, "OriginalClaims"] = -
year_results["OriginalClaims"]
    final_results.loc[year, "AdminCosts"] = -year_results["AdminCosts"]
    final_results.loc[year, "BondCosts"] = -year_results["BondCosts"]
    final_results.loc[year, "InterestIncome"] =
year_results["InterestIncome"]
    final_results.loc[year, "EndFunds"] = year_results["EndFunds"]
    return final_results

def convert_to_2025_qd(final_results: pd.DataFrame,
inflation_rate=INFLATION_RATE) -> pd.DataFrame:
    inflation_since_2025 = [(1+inflation_rate) ** x for x in range(10)]
    final_results = final_results.divide(inflation_since_2025,axis=0)
    final_results = final_results.drop(columns=["DamsRepaired",
"HazDamsRepaired"])
    final_results["EndFunds"] = final_results["EndFunds"] /
(1+inflation_rate)
    final_results = np.round(final_results, 2)
    return final_results

# Assumes constant rates for now, but you easily replace the interest_1_year,
inflation_rate, and climate_factors with series.

def simulate_10_year_program(
    dams: pd.DataFrame,
    admin_costs_m_q: float = INITIAL_ANNUAL_ADMIN_M_Q,
    repair_cost_matrix: np.ndarray = INITIAL_REPAIR_COSTS_M_Q,
    premium_z: float = PREMIUM_Z_SCORE,
    claim_z: float = CLAIM_Z_SCORE,
    bond_total_m_q: int = INITIAL_BOND_M_Q,
    bond_interest_rate: float = SPOT_10_YEAR,
    interest_1_year: float = SPOT_1_YEAR,
    inflation_rate: float = INFLATION_RATE,
    predicted_climate_effects: Dict[str, float] = CLIMATE_EFFECTS,
    true_climate_effects: Dict[str, float] = CLIMATE_EFFECTS,
    setup_cost_m_q: float = ESTABLISH_ADMIN_M_Q, ) -> Tuple[pd.DataFrame,
List[pd.DataFrame]]:
    # Preallocate

```

```

final_results = pd.DataFrame(data=np.zeros((10, 11), dtype="float64"),
index=np.arange(2025, 2035),
                                columns=["StartFunds", "Premium",
"RepairCosts", "DamsRepaired", "HazDamsRepaired", "Claims",
"OriginalClaims", "AdminCosts",
                                "BondCosts", "InterestIncome",
"EndFunds"])
yearly_dam_data = []
control_dams = dams.copy()

# Collect the bond
funds = bond_total_m_q

# Pay underwriting fee
funds = funds * (1 - BOND_UNDERWRITING_FEE)

# First year (2025)
print("Simulating 2025")
year_results = simulate_1_year(year_start_funds_m_q=funds, dams=dams,
control_dams=control_dams, repair_cost_matrix=repair_cost_matrix,
                                years_left=10, premium_z=premium_z,
claim_z=claim_z,
                                annual_admin_costs_m_q=admin_costs_m_q,
bond_total_m_q=bond_total_m_q,
                                bond_interest_rate=bond_interest_rate,
interest_1_year=interest_1_year,
                                inflation_rate=inflation_rate,
predicted_climate_effects=predicted_climate_effects,
true_climate_effects=true_climate_effects,
                                setup_cost_m_q=setup_cost_m_q,
close_bond=False)

update_final_results(final_results, 2025, starting_funds=funds,
year_results=year_results)

funds = year_results["EndFunds"]
repair_cost_matrix = year_results["RepairCostMatrix"]
dams = year_results["Dams"]
control_dams = year_results["ControlDams"]
admin_costs_m_q = year_results["NewAdminCosts"]
yearly_dam_data.append(dams.copy())

# Next 8 Years (2026-2033)
for year in range(2026, 2034):
    print(f"Simulating {year}")
    year_results = simulate_1_year(year_start_funds_m_q=funds, dams=dams,
control_dams=control_dams, repair_cost_matrix=repair_cost_matrix,
                                years_left=2035-year,
premium_z=premium_z, claim_z=claim_z,
                                annual_admin_costs_m_q=admin_costs_m_q, bond_total_m_q=bond_total_m_q,
                                bond_interest_rate=bond_interest_rate,
interest_1_year=interest_1_year,
                                inflation_rate=inflation_rate,
predicted_climate_effects=predicted_climate_effects,
true_climate_effects=true_climate_effects,
                                setup_cost_m_q=0, close_bond=False)

```

```

        update_final_results(final_results, year, starting_funds=funds,
year_results=year_results)

        funds = year_results["EndFunds"]
        repair_cost_matrix = year_results["RepairCostMatrix"]
        dams = year_results["Dams"]
        control_dams = year_results["ControlDams"]
        admin_costs_m_q = year_results["NewAdminCosts"]
        yearly_dam_data.append(dams.copy())
    # Final Year (2034)
    print("Simulating 2034")
    year_results = simulate_1_year(year_start_funds_m_q=funds, dams=dams,
control_dams=control_dams, repair_cost_matrix=repair_cost_matrix,
                                years_left=1,premium_z=premium_z,
claim_z=claim_z,
                                annual_admin_costs_m_q=admin_costs_m_q,
bond_total_m_q=bond_total_m_q,
                                bond_interest_rate=bond_interest_rate,
interest_1_year=interest_1_year,
                                inflation_rate=inflation_rate,
predicted_climate_effects=predicted_climate_effects,
true_climate_effects=true_climate_effects,
                                setup_cost_m_q=0, close_bond=True)
    update_final_results(final_results, 2034, starting_funds=funds,
year_results=year_results)

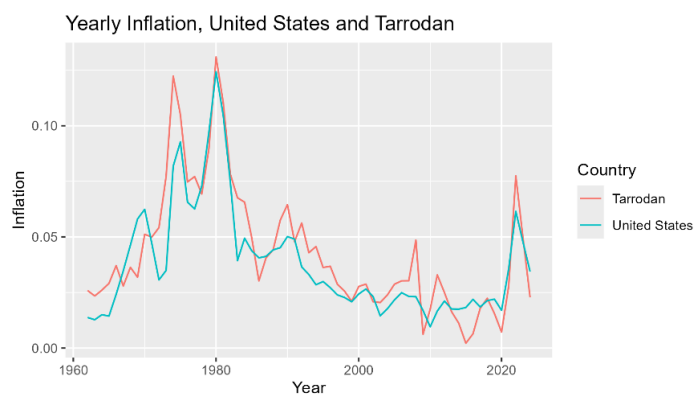
    dams = year_results["Dams"]
    yearly_dam_data.append(dams.copy())

    return final_results, yearly_dam_data

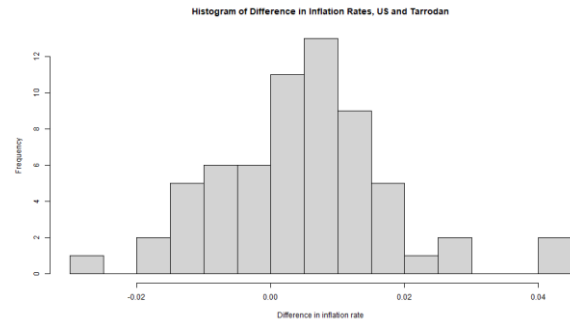
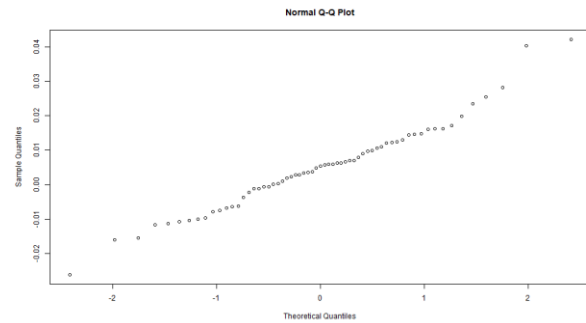
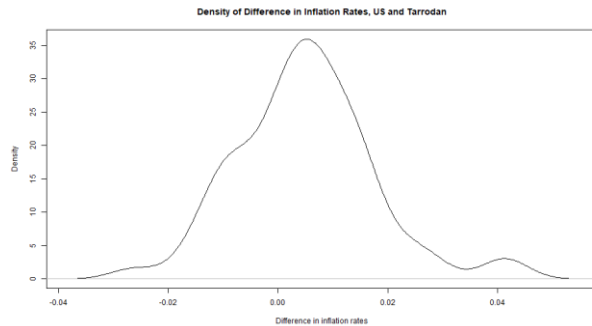
```

Appendix E – Inflation Analysis

The inflation rate of Tarrodon is of interest in order to approximate changes in costs over time. Not only might this affect interest rates and purchasing power, but also the cost of materials for dam renovations as well as adjusting worker salaries to handle the increasing cost of living. Previous year's inflation rates are available as part of the dataset. However, in order to ease the burden of forecasting a value that depends on many macroeconomic factors, it would be convenient to link this inflation rate to one that is more understood, such as the United States.



Firstly, in comparing the two rates, the difference between the two does not seem to be particularly significant. In fact, the difference between the two rates appears to be some sort of white noise. In order to verify this hypothesis, we first construct a histogram and a density plot on the difference in interest rates to see if a distribution appears.



Indeed, both graphs appear to indicate a normal distribution for the difference between interest rates, implying the Tarrodan interest rate simply differs from the United States interest rate by Gaussian white noise. To further confirm this, a normal QQ plot is created to compare the quantiles of the differences to the quantiles of the normal distribution. As most points appear in a straight line, there is no major evidence against this distribution being normal. Finally, we perform a Kolmogorov-Smirnov test to find a p-value comparing this distribution to the normal distribution. As we get $p=0.8645$, we do not have convincing evidence that this distribution is not normal. So, we will assume that it is.

Using this comparison, we are confident using the United States' predicted inflation rate as it will very likely be close to Tarrodan's. As such, we assume a forecasted inflation rate of 3% per year (Federal Reserve Bank of New York, 2025). We discuss forecasting best- and worst-case interest rates in Appendix G.

Appendix F – Stakeholder Analysis

Stakeholder	Interest	Power	Management Strategies
Citizens	Low 2	Low 1	Release information (renovations, floods), speak to press about plan
Dam Owners	High 4	Low 2	Directly inform (big weather, scheduled renovations/inspections, plan/pricing updates)
Businesses (general)	Low 2	Low 2	No special treatment than citizens

Businesses (dam-dependent)	High 5	High 3	Directly inform, treat well but not privileged (aid with impacts of maintenance)
Government	High 5	High 5	Hire inspectors, support staff, disaster relief personnel. Auditing reserves and premium collection.
Engineering Corps	High 5	Low 2	Get recommendations in Phase 1. Continue to update when more information is collected about dams and possible climate change escalation.
Construction Workers	High 4	Low 3	Collecting subcontractor bids and monitoring progress/overages.

The stakeholders of Tarrodan are its people, government, and businesses. The people of Tarrodan are split into three categories in respect to their wants as stakeholders. Since there are so many earthen dams in Tarrodan, there is a high likelihood that a given citizen lives near a dam or has a dam support their life in some notable way. The people of Tarrodan do not want their property or the locations they often use damaged by dam failure. Dam failure can also be catastrophic if the particular dam is used for necessary functions, such as irrigation or water supply. The people do not want dam failures to affect them, and they want to be compensated fairly for what they lose in the event of a failure.

Many of the earthen dams in Tarrodan are presumably on private land and privately owned. These people who own dams will need to pay premiums as mandated by their regional government. These citizens will want low premiums but also coverage for dam breakage its effects on property and the environment. The initial phase of the plan will be supported by bonds. The people who buy these bonds want to support their country in the effort to prevent catastrophes but also want a good return on their investment.

The government of Tarrodan wants its citizens, land, and environment protected while also avoiding deficit. The government needs to ensure that it takes in enough money for this program that is vital for protecting its citizens, infrastructure, and environment.

There are many ways that a business could be affected by dam failure. For example, failure may prevent a business' employees and customers from physically getting to the business location or mar scenic views that are necessary for the well-being of the tourism industry. Businesses have a lot to lose in the event of a dam failure, and they want fair reimbursement if they are in any way impacted. This plan will provide that desired reimbursement as well as reduce the probability of failure for many important dams around the country, further reducing the impact of failure on these businesses.

Appendix G – Financial Calculations

G.1 – Repair Bond

In order to fund the repairs for this program, we will take on ten-year bonds at the start of the program, using the savings from repairs to pay off the bond at the end of the period. In particular, we expect repair costs of 26,981Qm in 2025 Qalkoons, which is then the price of our proposed bond. The ten-year spot rate is then approximated using an ARIMA(0,1,1) model on the provided 1962-2024 data, predicting a 2025 ten-year spot rate of 4.221%. Of course, when this plan is implemented, the true value of this rate will be known. We also assume that bond coupons will be equal to the interest rate, paid yearly, for simplicity.

Thus, for a bond of price 26,906Qm with yearly interest 4.221% and equal yearly coupons, we expect to pay 1,130Qm yearly and an additional 26,906Qm at the end of year ten. We further expect a 1.07% underwriting fee (Joffe, 2015), which is deducted from our starting fund calculation.

G.2 – Premium Calculations

This program collects all its funding for claims from the respective regional governments. In order to equitably source these funds, the amount each region is responsible for is proportional to the GDP of each region as of 2023. These figures are 4,197Qb for Flumevale, 480Qb for Lyndrassia, and 3,396Qb for Navaldia, or 51.98%, 5.94%, and 42.07% respectively.

To guarantee solvency, we collect premiums equal to the 95th percentile of expected claims, updated yearly due to repairs and climate change. For the first year, we collect 95,207Qm, which is then split as 49,495Qm from Flumevale, 5,662Qm from Lyndrassia, and 40,049Qm from Navaldia. The full prediction of total premiums is seen in Table 1, not split for GDP. The percentage that each region is expected to contribute is not adjusted yearly. The GDP data that was provided has very few data points, two of which are COVID and the subsequent rebound, which are major outliers. Due to both of these factors, it is difficult to reliably estimate regional GDP growth, so we do not factor this into analysis. In the event of major events that significantly alter the GDP split of the nation, or in the presence of more accurate GDP trend predictions, this can be re-evaluated at the program management's discretion.

G.3 – Labor Costs

Labor costs are a notable expense of this program. Firstly, all dams will have a mandated inspection at least once every five years. With 20,806 dams over 5 years, an estimated 1 inspection per week, and 50 work weeks per year, we require at least $20806 / (5 * 50 * 1) = 84$ inspectors. To guarantee success, we plan to employ 120 total inspectors, each earning a salary of 100,000Q – approximately the United States dam engineer median salary – with benefits of around 60,000Q, which is 40% of the total compensation (Bureau of Labor Statistics, 2024). In total, this requires $160,000 * 120 = 19.2$ Qm at a minimum of labor costs per year. For simplicity, we then estimate

yearly costs at 2.5 times that value to include support staff, management, and other employees, giving 50Qm yearly.

We also require a few one-time expenses at the beginning of the program. These include office space, computers and other technology, and various smaller expenses to fully establish the support staff to run this program. There is very little existing data about the costs of establishing a government agency that is available to the public. As such, we make a rough estimate of 2 years of labor costs, giving 100Qm in the first year of set-up expenses.

Appendix H – Sensitivity Analysis

As is standard, sensitivity analysis is conducted by adjusting the value of a single factor and holding the other factors at their expected value. The factors that are of interest are as follows: inflation rate, one-year interest rate, repair costs, and climate change.

Inflation rate is notoriously variable, and it is important to ensure our program remains productive even against high inflation. We expect inflation of 3%, which is the projected value for the United States, as justified in Appendix E. The best- and worst-case estimates are the minimum and maximum ten-year inflation rates for Tarrodan’s historical data. These provide a reasonable estimate of what Tarrodan’s macroeconomic situation may become but are not heavily influenced by unusually high or low spikes that are not likely to last more than one or two years.

Interest rates are similarly difficult to estimate. Because our ten-year bond is taken at the start of the program, the only interest rate of concern refers to how much the program’s excess can generate without locking too much of our reserves in investments and being unable to pay unusually high claims. We expect a one-year spot rate interest rate of 5.9%, which is the average of ten-year rates in Tarrodan’s historical data. In the best case, we consider 12% interest, which is the maximum ten-year rate. As for the worst case, it is possible that none of the funds can be invested for a variety of reasons, so we use 0% interest.

Repair costs are unlikely to be lower than predicted due to the nature of the study providing these figures. Expected costs are clearly those given by that study (Association of State Dam Safety Officials, 2025). However, it is of reasonable concern that repair costs are nearly 2.2 times higher than predicted (Petheram & McMahon, 2019).

Climate change is very difficult to estimate. The method used, as described in Appendix D, has logical basis but not scientific, as there is no existing literature to draw from. Thus, it is highly important to vary the factors given to provide a chance of capturing the actual outcome. For this, we simply halve or double the factors at each level.

High inflation

	PREMIUM	REPAIRS	CLAIMS	ADMIN	BOND	INTEREST	END FUNDS
2025	95207.29	-871.65	-79754.72	-150.0	-1130.06	2355.23	38666.34
2026	92746.30	-977.65	-78411.66	-50.0	-1033.61	3005.44	49341.15
2027	91254.68	-954.80	-77472.51	-50.0	-945.40	3609.21	59253.40
2028	90218.56	-1124.62	-76661.21	-50.0	-864.71	4175.51	68550.49
2029	89323.99	-1361.27	-75965.03	-50.0	-790.91	4702.73	77205.91
2030	88554.70	-1684.56	-75307.31	-50.0	-723.41	5191.73	85233.87
2031	87832.45	-1820.54	-74849.70	-50.0	-661.67	5645.38	92681.67
2032	87333.36	-2018.82	-74484.35	-50.0	-605.20	6068.54	99628.84
2033	86935.61	-2216.46	-74326.97	-50.0	-553.55	6455.63	105983.76
2034	86768.51	-9086.65	-73616.98	-50.0	-12561.13	5748.81	94379.76

Low inflation

	PREMIUM	REPAIRS	CLAIMS	ADMIN	BOND	INTEREST	END FUNDS
2025	95207.29	-871.65	-79754.72	-150.0	-1130.06	2355.23	41622.49
2026	92746.30	-977.64	-78411.65	-50.0	-1112.64	3175.20	56113.32
2027	91254.68	-954.80	-77472.51	-50.0	-1095.48	3999.92	70688.14
2028	90218.56	-1124.62	-76661.21	-50.0	-1078.59	4837.54	85491.04
2029	89323.99	-1361.27	-75965.03	-50.0	-1061.96	5686.23	100489.34
2030	88554.69	-1684.56	-75307.31	-50.0	-1045.59	6546.44	115691.29
2031	87832.45	-1820.54	-74849.70	-50.0	-1029.46	7420.67	131141.03
2032	87333.36	-2018.82	-74484.35	-50.0	-1013.59	8313.55	146920.42
2033	86935.61	-2216.46	-74326.97	-50.0	-997.96	9219.62	162932.72
2034	86768.51	-9086.65	-73616.97	-50.0	-24377.18	8411.65	148654.17

No interest

	PREMIUM	REPAIRS	CLAIMS	ADMIN	BOND	INTEREST	END FUNDS
2025	95207.29	-871.65	-79754.72	-150.0	-1130.06	0.0	38756.38
2026	92746.30	-977.64	-78411.66	-50.0	-1097.15	0.0	49481.79
2027	91254.68	-954.80	-77472.51	-50.0	-1065.19	0.0	59411.62
2028	90218.55	-1124.63	-76661.21	-50.0	-1034.16	0.0	68699.20
2029	89323.99	-1361.27	-75965.04	-50.0	-1004.04	0.0	77323.16
2030	88554.69	-1684.56	-75307.32	-50.0	-974.80	0.0	85302.12
2031	87832.45	-1820.54	-74849.70	-50.0	-946.41	0.0	92687.30
2032	87333.36	-2018.82	-74484.35	-50.0	-918.84	0.0	99561.81
2033	86935.61	-2216.46	-74326.97	-50.0	-892.08	0.0	105836.80
2034	86768.51	-9086.65	-73616.97	-50.0	-21487.38	0.0	85790.59

High interest

	PREMIUM	REPAIRS	CLAIMS	ADMIN	BOND	INTEREST	END FUNDS
2025	95207.29	-871.65	-79754.72	-150.0	-1130.06	4790.29	43407.15
2026	92746.30	-977.64	-78411.66	-50.0	-1097.15	6674.04	60476.75
2027	91254.68	-954.80	-77472.51	-50.0	-1065.19	8662.67	78496.70
2028	90218.55	-1124.63	-76661.21	-50.0	-1034.16	10781.43	97695.81
2029	89323.99	-1361.27	-75965.04	-50.0	-1004.04	13036.73	118132.23
2030	88554.69	-1684.56	-75307.32	-50.0	-974.80	15440.43	139913.29
2031	87832.45	-1820.54	-74849.70	-50.0	-946.41	18009.49	163192.79
2032	87333.36	-2018.82	-74484.35	-50.0	-918.84	20766.50	188175.39
2033	86935.61	-2216.46	-74326.97	-50.0	-892.08	23715.06	214893.73
2034	86768.51	-9086.65	-73616.97	-50.0	-21487.38	23690.55	214671.64

Repair cost high

	PREMIUM	REPAIRS	CLAIMS	ADMIN	BOND	INTEREST	END FUNDS
2025	95207.29	-1195.77	-80264.65	-150.0	-1130.06	2306.02	40185.48
2026	93337.57	-1447.05	-79340.38	-50.0	-1097.15	3043.72	53040.97
2027	92319.25	-1703.59	-78743.38	-50.0	-1065.19	3764.09	65594.33
2028	91663.56	-2175.63	-78278.41	-50.0	-1034.16	4467.47	77851.60
2029	91141.98	-3494.95	-77279.85	-50.0	-1004.04	5142.72	89618.89
2030	90026.55	-3737.94	-76647.42	-50.0	-974.80	5795.88	101001.12
2031	89323.75	-3977.97	-76197.22	-50.0	-946.41	6440.04	112226.53
2032	88823.34	-4400.38	-75807.20	-50.0	-918.84	7072.53	123248.53
2033	88395.56	-5009.32	-75451.17	-50.0	-892.08	7684.25	133908.51
2034	88018.56	-20589.52	-74337.73	-50.0	-21487.38	6222.28	108431.7

Severe climate change

	PREMIUM	REPAIRS	CLAIMS	ADMIN	BOND	INTEREST	END FUNDS
2025	95207.29	-871.65	-79754.72	-150.0	-1130.06	2355.23	41043.01
2026	92746.30	-977.64	-78614.23	-50.0	-1097.15	3129.97	54543.94
2027	91254.68	-954.80	-77848.82	-50.0	-1065.19	3886.91	67734.69
2028	90218.55	-1124.63	-77196.11	-50.0	-1034.16	4634.35	80759.90
2029	89323.99	-1361.27	-76640.85	-50.0	-1004.04	5370.64	93590.66
2030	88554.69	-1684.56	-76108.66	-50.0	-974.80	6096.32	106236.55
2031	87832.45	-1820.54	-75768.60	-50.0	-946.41	6813.52	118734.93
2032	87333.36	-2018.82	-75505.30	-50.0	-918.84	7526.94	131167.27
2033	86935.61	-2216.46	-75437.48	-50.0	-892.08	8230.90	143434.72
2034	86768.51	-9086.65	-74733.88	-50.0	-21487.38	7365.87	128360.38

Mild climate change

	PREMIUM	REPAIRS	CLAIMS	ADMIN	BOND	INTEREST	END FUNDS
2025	95207.29	-871.65	-79754.72	-150.0	-1130.06	2355.23	41043.01
2026	92746.30	-977.64	-78310.36	-50.0	-1097.15	3147.89	54856.37
2027	91254.68	-954.80	-77285.43	-50.0	-1065.19	3938.58	68635.17
2028	90218.55	-1124.63	-76396.81	-50.0	-1034.16	4734.64	82507.53
2029	89323.99	-1361.27	-75632.96	-50.0	-1004.04	5533.21	96423.76
2030	88554.69	-1684.56	-74915.94	-50.0	-974.80	6333.83	110375.72
2031	87832.45	-1820.54	-74403.75	-50.0	-946.41	7138.26	124393.92
2032	87333.36	-2018.82	-73992.06	-50.0	-918.84	7950.11	138541.43
2033	86935.61	-2216.46	-73794.99	-50.0	-892.08	8762.88	152705.23
2034	86768.51	-9086.65	-73085.86	-50.0	-21487.38	8010.07	139586.33

Appendix I – Future Investments

There are several programs that have been recommended by the scientific community when it comes to dam efficiency. AES has combed scientific journals and found the following programs that have the highest probability of increasing dam safety measures.

Studies have shown that many dams have the unfortunate combination of negative ecological impact as well as having repair costs exceeding their benefits (Ho et al., 2017). Excess funds from premium collection should be used towards analyzing existing dams for cost savings measures related to removing unnecessary dams. Part of the analysis of dam removal will need to go towards sediment analysis and the environmental impact from newly freed waterways (Concha Larrauri & Lall, n.d.). Sedimentation is shown to be a leading cause of overtopping a dam, which is when a dam does not break, but instead overflows and causes some degree of flooding. Scientists have called for more studies on the rate of buildup and optimal sedimentation clearance methods, of which our program also recommends putting excess reserves towards.

Flood zones are consistently being reassessed due to the impact of climate change on local waterways and coastal areas (Banham, 2024). The flood map evaluation criteria in the United States have been recently updated, which translated into increased likelihood of flooding in existing areas, as well as newly designated flood areas. Tarrodan will likely need to analyze their flood mapping process and issue updated maps.

Climate change is the primary factor for the rising water levels globally. Dam heights that were previously rated as satisfactory may change as rising water levels increase their risk factor. As the engineers update ratings during the 5-year mandatory inspection period, they may collectively change the requirements for defining a satisfactory dam, which will change the existing ratings

and require further dam renovations. Excess reserve funds should be allocated towards this new round of renovations.

The research done as a part of these programs should be used to inform Tarrodan's policies, laws, and codes relating to dams, so that in the future the citizens of Tarrodan will be better protected from dam failure.